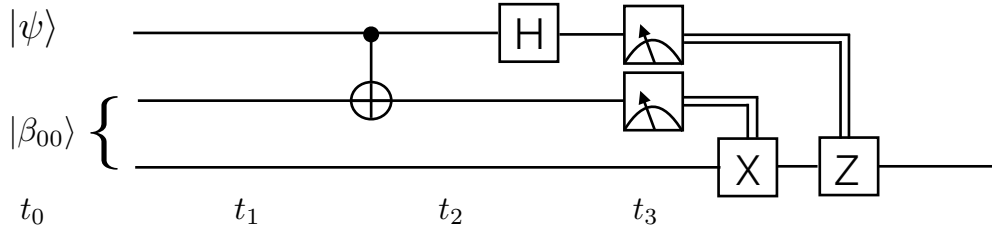


## 4.2 Teleportation (Bennet et al. 1993)

The teleportation protocol allows Alice to “send” an unknown qubit  $|\psi\rangle$  to Bob without using a quantum channel, i.e. without physically sending the qubit, but using only a classical channel, as for example a phone communication. To achieve this, Alice and Bob share a pair of qubits in the entangled state  $|\beta_{00}\rangle = \frac{|00\rangle + |11\rangle}{\sqrt{2}}$ . Alice entangles her qubit with the unknown qubit  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$  she wants to teleport, using a CNOT gate. She then applies the Hadamard gate to the first qubit (the qubit that originally was in the state  $|\psi\rangle$ ) and measures the two qubits. After this, Alice telephones the result of her measurements to Bob. According to the result 00, 01, 10, 11 communicated by Alice, Bob effects one of four operations on his qubit, respectively  $I, Z, X, ZX$ , and by so doing transforms the state of his qubit exactly in the state  $|\psi\rangle$  of the qubit originally owned by Alice.

The corresponding quantum circuit is



**Fig. 4.1** Teleportation circuit

a 3-qubit circuit where the upper two qubits belong to Alice and the lower qubit belongs to Bob. By following the evolution of the initial 3-qubit state  $(\alpha|0\rangle + \beta|1\rangle)\frac{|00\rangle + |11\rangle}{\sqrt{2}}$  across the CNOT and H gates, one finds that the 3-qubit state before Alice measurements is

$$\frac{1}{2} [|00\rangle(\alpha|0\rangle + \beta|1\rangle) + |01\rangle(\alpha|1\rangle + \beta|0\rangle) + |10\rangle(\alpha|0\rangle - \beta|1\rangle) + |11\rangle(\alpha|1\rangle - \beta|0\rangle)] \quad (4.1)$$

A measurement by Alice produces a collapse into one of the four states contained in this superposition. For example if Alice measures 01 the state (4.1) collapses into

$$|01\rangle(\alpha|1\rangle + \beta|0\rangle) \quad (4.2)$$

This state is a product state, with Bob’s qubit in the state

$$\alpha|1\rangle + \beta|0\rangle \quad (4.3)$$

Alice phones her result 01 to Bob, so that Bob learns that his qubit is in the state (4.3). He therefore applies to it the gate X and reconstructs the original  $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$  state.

• **Observations:**

- i) the state  $|\psi\rangle$  is teleported at a speed always  $\leq c$ , since it is limited by the speed of the classical communication.
- ii) the original state  $|\psi\rangle$  owned by Alice gets destroyed (by measurement), so that there is no violation of the no cloning theorem.
- iii) teleportation is not vulnerable to noise, since it does not use a physical carrier. (The classical part of the protocol, i.e. communication of Alice's results to Bob, could be disturbed by classical causes).

## 5 Lecture 5

### 5.1 Quantum parallelism

Quantum gates implement linear operations, and therefore can operate in parallel on all the states contained in a superposition. Consider a Boolean function  $f$ , i.e. a function taking  $N$  bits into 1 bit. Thus the domain of  $f$  is the set of binary  $N$ -bit integers  $x$  and the image is  $\{0, 1\}$ . By  $|x\rangle$  we denote the  $N$ -qubit state specified by the string of 0's and 1's in  $x$ : thus for  $N=4$  if  $x = 0110$ , then  $|x\rangle = |0\rangle|1\rangle|1\rangle|0\rangle$ . We define now a unitary  $N+1$  qubit operator  $U_f$  that acts on the computational basis of  $N+1$  qubits as follows:

$$U_f(|x\rangle|y\rangle) = |x\rangle|y \oplus f(x)\rangle \quad (5.1)$$

where  $\oplus$  denotes addition modulo 2. If the input  $N+1$  qubit state is a “democratic” superposition of all the states of the computational basis, tensored with  $|0\rangle$ , we find

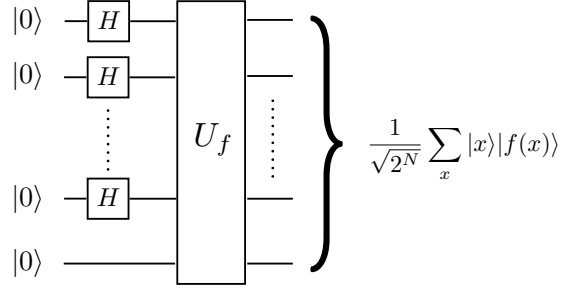
$$U_f\left(\frac{1}{\sqrt{2^N}} \sum_x |x\rangle|0\rangle\right) = \frac{1}{\sqrt{2^N}} \sum_x |x\rangle|f(x)\rangle \quad (5.2)$$

We see that applying  $U_f$  only once has produced a state containing *all the values* of the function  $f$ , an operation that classically would require  $2^N$  evaluations of the function  $f$ , one for each value of the variable  $x$ . But extracting this information from the output state is not so easy. Measuring the  $N+1$  qubit output state in the computational basis yields a particular value for  $x$ , giving the corresponding value for  $f(x)$ , and destroys all the rest of the information (all the other  $x, f(x)$  values). We will see in next Section how interference may be used to recover more information.

The superposed state  $\frac{1}{\sqrt{2^N}} \sum_x |x\rangle$  can be easily obtained by using  $N$  Hadamard gates on  $N$  qubits in the state  $|0\rangle$ :

$$(H \otimes H \otimes \dots \otimes H)|0\rangle|0\rangle\dots|0\rangle = \frac{1}{\sqrt{2^N}}(|0\rangle + |1\rangle)(|0\rangle + |1\rangle)\dots(|0\rangle + |1\rangle) = \frac{1}{\sqrt{2^N}} \sum_x |x\rangle \quad (5.3)$$

The combined action of the Hadamard gates and the  $U_f$  operator is represented by the following circuit:

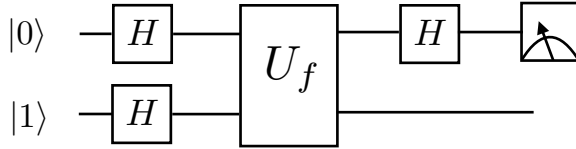


**Fig. 5.1** Parallel evaluation of a Boolean function  $f$

- **Exercise:** prove that  $U_f$  is unitary.
- **Exercise:** find the circuit that implements  $U_f$  when  $N = 1$ .

## 5.2 Deutsch algorithm

This algorithm allows to find whether a function  $f$  is constant or nonconstant, by combining parallelism and interference. Consider the circuit:



**Fig. 5.2** Circuit for Deutsch's algorithm  $f$

where  $U_f$  is defined in (5.1). Following the evolution of the initial 2-qubit state  $|0\rangle|1\rangle$  through the unitary gates yields as final state:

$$\frac{|0\rangle|f(0)\rangle - |0\rangle|1 \oplus f(0)\rangle}{\sqrt{2}} = \pm |0\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad \text{if } f(0) = f(1) \quad (5.4)$$

$$\frac{|1\rangle|f(0)\rangle - |1\rangle|f(1)\rangle}{\sqrt{2}} = \pm |1\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad \text{if } f(0) \neq f(1) \quad (5.5)$$

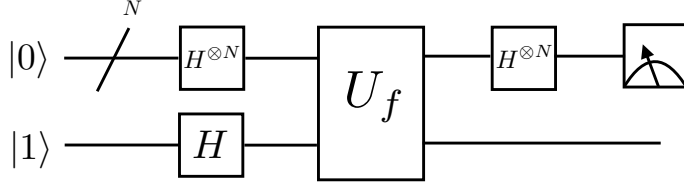
or in a single formula:

$$\pm |f(0) \oplus f(1)\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (5.6)$$

Thus by measuring the first qubit we can determine whether the function  $f$  is constant (result 0) or nonconstant (result 1). Constancy of  $f$  is a global property, and classically two evaluations of  $f$  would be necessary to establish it, while a single application of Deutsch's quantum circuit is sufficient.

### 5.3 Deutsch-Jozsa algorithm

Combining together the two preceding Sections leads to a generalization of the Deutsch algorithm, due to Deutsch and Jozsa. The circuit is:



**Fig. 5.3** Circuit for Deutsch-Jozsa algorithm  $f$

where the slash in the first line indicates  $N$  qubits in the first register. The input state is a  $N + 1$  qubit state  $|0\dots 0\rangle|1\rangle$ , with  $|0\dots 0\rangle =$  the state of  $N$  qubits all initialized to  $|0\rangle$ . The  $N + 1$  qubit state emerging from the first battery of  $H$  gates is

$$\frac{1}{\sqrt{2^N}} \sum_{x=0}^{2^N-1} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (5.7)$$

which becomes, after applying  $U_f$ :

$$\frac{1}{\sqrt{2^N}} \sum_{x=0}^{2^N-1} |x\rangle \frac{|f(x)\rangle - |1 \oplus f(x)\rangle}{\sqrt{2}} = \frac{1}{\sqrt{2^N}} \sum_{x=0}^{2^N-1} (-1)^{f(x)} |x\rangle \frac{|0\rangle - |1\rangle}{\sqrt{2}} \quad (5.8)$$

Observe now that the action of  $H^{\otimes N}$  on  $|x\rangle = |x_1\rangle|x_2\rangle\dots|x_N\rangle$ :

$$H^{\otimes N} |x\rangle = \frac{1}{\sqrt{2^N}} (|0\rangle + (-1)^{x_1}|1\rangle) (|0\rangle + (-1)^{x_2}|1\rangle) \dots (|0\rangle + (-1)^{x_N}|1\rangle) \quad (5.9)$$

can also be written as

$$H^{\otimes N} |x\rangle = \frac{1}{\sqrt{2^N}} \sum_{y=0}^{2^N-1} (-1)^{x \cdot y} |y\rangle \quad (5.10)$$

with the definition

$$x \cdot y \equiv x_1 y_1 \oplus x_2 y_2 \oplus \dots \oplus x_N y_N \quad (5.11)$$

Then the state of the first  $N$ -qubit register before the measurement is

$$\frac{1}{2^N} \sum_{y=0}^{2^N-1} \left[ \sum_{x=0}^{2^N-1} (-1)^{f(x)} (-1)^{x \cdot y} \right] |y\rangle \quad (5.12)$$

Thus the probability of finding  $0, 0, \dots, 0$  in a measurement of the first register ( $N$  qubits) is

$$p(0, 0, \dots, 0) = \left| \frac{1}{2^N} \sum_{x=0}^{2^N-1} (-1)^{f(x)} \right|^2 \quad (5.13)$$

This probability is 1 if  $f(x)$  is *constant*, and is 0 if  $f(x)$  is *balanced*, i.e. with an equal number of values 1 and 0. Then by measuring the first register ( $N$  measurements) we can find whether  $f$  is constant or not. The total number of operations grows linearly with  $N$ , even considering the  $H$  gates in the circuit 5.3, and only one “run” of  $U_f$  is necessary.

By contrast, to obtain this information classically would require  $2^N$  evaluations of the function  $f$ , growing exponentially with  $N$ .